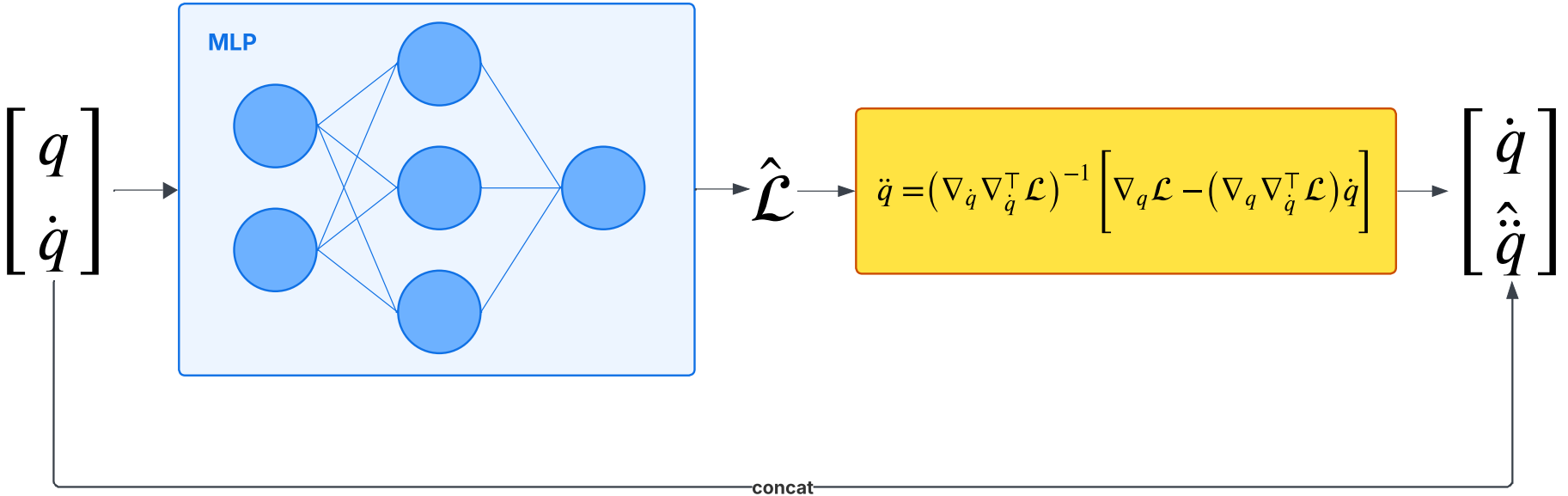
MLP
$\begin{bmatrix} q \\ \dot{q} \end{bmatrix}$
$\hat{\mathcal{L}}$
$\ddot{q} = \left(\nabla_{\dot{q}} \nabla_{\dot{q}}^{\top} \mathcal{L}\right)^{-1} \left[\nabla_q \mathcal{L} - \left(\nabla_q \nabla_{\dot{q}}^{\top} \mathcal{L}\right) \dot{q}\right]$
$\begin{bmatrix} \dot{q} \\ \hat{\ddot{q}} \end{bmatrix}$
concat